Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Branch: B.Tech Honors Computer Science and Engineering (Artificial Intelligence) **Semester - VII**

Subject: Intelligent System and Robotics Lab **Subject Code: D127791(022)**

Maximum Marks in ESE: 40

List of Experiments (Any 10)

1. Path Planning Algorithms: Implement algorithms such as A*, Dijkstra, or RRT to find the optimal path for a robot to navigate in a known environment.
2. PID Control: Design a PID controller to regulate the position or velocity of a robot arm or mobile robot.
3. SLAM (Simultaneous Localization and Mapping): Implement SLAM algorithms like EKF SLAM or FastSLAM for mapping and localization in robotics.
4. Reinforcement Learning for Robot Navigation: Train a robot to navigate through a maze or reach a target using reinforcement learning algorithms like Q-learning or Deep Q-Networks.
5. Object Detection and Recognition: Use deep learning frameworks like Tensor Flow or PyTorch to detect and recognize objects in images or video streams captured by a robot's camera.
6. Voice Command Recognition: Develop a system to recognize and respond to voice commands using speech recognition libraries such as Speech Recognition in Python.
7. Gesture Recognition: Implement a gesture recognition system using computer vision techniques like Open CV to interpret hand gestures and control a robot's movements.
8. Swarm Robotics Simulation: Simulate a swarm of robots using tools like ROS (Robot Operating System) or V-REP to study collective behaviours and coordination strategies.
9. Neural Network-based Robot Control: Train neural networks to control the movement or manipulation of a robotic arm or gripper using frameworks like Keras or Tensor Flow.
10. Evolutionary Robotics: Use evolutionary algorithms like Genetic Algorithms or Genetic Programming to evolve robot controllers for specific tasks.
11. Robot Localization with Kalman Filters: Implement Extended Kalman Filters (EKF) or Unscented Kalman Filters (UKF) for robot localization using noisy sensor data.
12. Mobile Robot Simulation: Simulate the movement and interaction of mobile robots in different environments using tools like Gazebo or Webots.
13. Behavior-Based Robotics: Design and implement behaviors for a robot using a behavior-based approach, combining simple reactive behaviors to achieve complex tasks.
14. Human-Robot Interaction: Develop interfaces for humans to interact with robots, such as speech interfaces, gesture interfaces, or haptic feedback using Python libraries or Matlab GUI tools.
15. Robot Soccer Simulation: Create a simulated robot soccer environment using platforms like Robo Cup Soccer Simulation or Pygame to implement robot strategies and compete against other simulated teams.

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Branch: B.Tech Honors Computer Science and Engineering (Artificial Intelligence) **Semester - VII**

Subject: Technical Communication and Soft Skill **Subject Code: D127001(022)**

Maximum Marks (TA): 20

1. **Effective Communication in AI**
   Principles of effective technical communication. Written and verbal communication skills. Presentation techniques for conveying AI concepts.

2. **Teamwork and Collaboration**
   Importance of teamwork in AI projects. Collaboration tools and techniques.

3. **Problem-Solving Skills**
   Critical thinking and problem-solving strategies. Analytical skills for AI applications.

4. **Time Management and Productivity**
   Time management techniques for project completion. Prioritization and task management skills.

5. **Professional Ethics and Integrity**
   Ethical considerations in AI research and development. Maintaining integrity and professionalism in the workplace.